

Probabilistic Plug Formation

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1 Introduction

2 Model

Brief description of model equation

3 Results

Things to show:

- Location and width of ‘transition region’ for various values of ‘la’ and ‘S’
- PDF fitted to high ‘la’ curves?

Figure 1 shows the critical thickness parameter a_c as a function of Bo . For fixed Bo and $a > a_c$, plugs will form; for $a < a_c$, plugs may form, but only if a is ‘close enough’ to a_c (where ‘close enough’ depends on Bo and tube length).

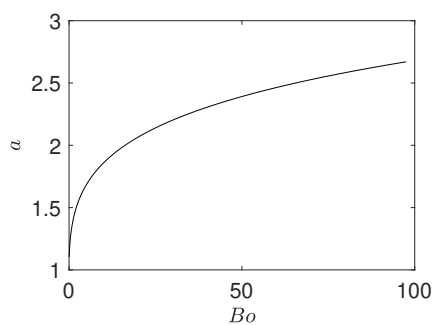


Figure 1: Critical value a_c as a function of each value of Bo . For fixed Bo and for $a > a_c$, plugs will form.

3.1 Toy model for plug formation?

Let N be the number of wave crests being tracked. This should probably be proportional to the domain length, perhaps $N \approx L/(2\sqrt{2}\pi)$ where L is the domain length? Something else?

We will define a system of $2N$ ODEs, with N equations tracking the location (in z) of each wave, and N equations tracking the amplitude of each wave (or really the distance R of each wave crest from the center of the tube).

Here is a best guess as to what these equations might look like; for $j = 1, \dots, N$:

$$\dot{x}_j = -\frac{1}{2} \left(1 - \frac{a^2}{R_j^2} + 2 \ln \frac{a}{R_j} \right), \quad (1a)$$

$$\dot{R}_j = (R_j - 1)(R_j - R_b)(R_j - R_c)\omega - (R_j - 1) \sum_{i \neq j} s(R_i, R_j, X_i, X_j; p) (|R_i - R_j| - p) c(R_i, R_j) \quad (1b)$$

where R_b is the expected ‘amplitude’ of a wave for a given thickness parameter a from the **traveling wave family** curve; R_c is the **wave amplitude corresponding to the critical thickness a_c in the bifurcation curve**; p is a threshold parameter that will have to be chosen; $\omega(R_j)$ is the linear growth rate of waves predicted by linear stability analysis and is given by (something like):

$$\omega = \frac{a^2}{64Bo(1 - R_c)(1 - R_b)} (a^4 + 3 - 4a^2 + 4 \ln a) \quad (2)$$

and $c(R_i, R_j)$ defines the strength of the interaction between waves i and j and might have functional form something like

$$c(R_i, R_j) = \lambda_1 e^{-\lambda_2 d(x_j, x_i)^2} \quad (3)$$

where $d(x_j, x_i)$ is a measure of the distance between waves j and i keeping in mind the periodic boundary conditions.

And where s is a sign function that takes on 1 or -1 depending on the amplitude and location of the waves: if $|R_j - R_i| > p$, then ... fill in from Notion account notes...

Equations (1a) will give the location in z of each wave; the wave speed is being approximated here by the linear wave speed predicted by linear stability analysis. (We could try to make this more accurate numerically using the traveling wave family data, but this might suffice). Equations (1b) describe the amplitude of the wave (well, $0 < R < 1$ represents the distance of the wave from the center of the tube, so small R means big wave, and R close to 1 means small wave). The first term in equation (1b) represents logistic growth of the wave, beginning as a small-amplitude wave (R close to 1 initially) and tending towards R_b over time, with R_b determined numerically from bifurcation curve data. The summation in equation (1b) is meant to describe the interactions

between waves i and j ; there should be virtually no interaction if the waves are far apart from each other (hence the Gaussian factor c). If waves i and j are close to one another, then they will interact, with the difference in amplitude between the waves decreasing if the difference is smaller than p , and increasing if the difference is larger than p . (This part will need to be worked out a bit...)

This is deterministic; we will likely want to add noise here. But where? I suspect either (i) adding (white?) noise to equation (1b) or (ii) adding (white?) noise within the interaction factor c might be a good starting point. Not sure though!

For starters, we'll want to code up and simulate this system. For initial conditions, use R_j smaller than but close to 1; I'd use random numbers between 0.98 and 1, e.g. For initial values of x_j , I'd use evenly spaced values from 0 to L . For the parameters, let's try $p = 0.1$ (something in the range of 0.01 up to 0.2, probably closer to 0.01 than to 0.2 is my best guess). No idea for the $\lambda_{1,2}$ values, we might have to experiment with those. For R_b , this can be read right off the bifurcation data I sent a while back in Matlab. For λ_2 , we want it to be approximately 1 over the width of a wave (maybe $1/\pi$?) So maybe try $\lambda = 1/3^2$ or $1/4^2$?

Note: the equations aren't quite right above – I haven't taken into account periodic BCs! So when x_j reaches the end of the tube, i.e. $x_j = L$, it should reset to $x_j = 0$. And in equation (3), the distance $x_j - x_i$ should be a 'modular' distance, i.e., if $L = 50$ and $x_j = 45$ and $x_i = 2$, this distance should be 7.

Please note that it's also possible there are sign mistakes and typos!

3.2 Update on 8/21/2025

Let's use $\lambda_1 = 0$ for now to make sure the rest of the code is working as we expect. Let's use $R_b = 0.75$, and $R_c = 0.7$ for now.

3.3 Update on 12/18/2025

Here's one simple-minded approach to estimating parameters that might be good enough to demonstrate 'proof-of-concept' for this ODE model:

- Set $p = 0.05$ and $\lambda_2 = 1/\pi^2$ (or $1/\pi$? depending on whether it's d or d^2 that we use..., pretty sure it's d^2 .) Set $Bo = 1$ (keep fixed for all tests).
- Reed will supply values of R_c (which only depends on Bp so will be a constant for all tests here) and values of R_b (which will vary with a).
- Choose an a value where roughly half the time plugs form in the PDE. Run a bunch of ODE simulations with different initial conditions for one value of λ_1 ; then repeat with other values of λ_1 as needed. Choose λ_1 so that roughly half the ODE simulations form plugs.
- Next, calculate a distribution of R_j values (ignoring the initial transient (spin-up) times) for any PDE simulations for which plugs do not form.

Calculate the same thing for the ODE simulations. Check that they have roughly the same mean (should be approximately R_b , right?) and variance/standard deviation. If they are wildly different, then either try (i) a slightly different value of λ_1 or (ii) slightly different values of p or λ_2 (but stay within the ranges described above, namely $p \in (0.01, 0.2)$ and $\lambda_2 \in (1/5, 1/25)$ (don't feel too confident about this range!)).

- Last, using the values that you found for the first value of a you chose, now repeat the ODE simulations with a few different values of a that are in the ‘intermediate thickness’ range where plugs sometimes form. Check to see how well the ODE model predicts the percentage of time plugs form for these other values. We may only get qualitative agreement, but hopefully we can get something that's not too far off from the nice plots from the PDE model that you've already made.